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Amorphous SiO₂ surface models: energetics of the dehydroxylation process, strain, *ab initio* atomistic thermodynamics and IR spectroscopic signatures†

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In this contribution, realistic amorphous SiO₂ models of 2.1 × 2.1 nm with silanol densities ranging 1.1–7.2 OH per nm² are obtained by means of *ab initio* calculations via the dehydroxylation of a fully hydroxylated silica surface. The dehydroxylation process is considered to take place via direct condensation of adjacent silanol groups and silica migration steps. The latter reconstructions are needed in order to obtain highly dehydroxylated silica surfaces with favorable energetics and without the formation of defects. The obtained surface phase diagram of different silica models as a function of temperature and $P_{\text{H}_2\text{O}}$ is able to correctly describe the silanol density under different conditions, and the IR spectroscopic signatures of the silanols are in qualitative agreement with the experiment. The amorphous silica models presented here have a high degree of heterogeneity as found from the big variability obtained in the energetics of the dehydroxylation steps. It was also found that the resulting average Si–O distance of the newly formed siloxane bridges serves as a descriptor of the strain introduced in the silica surface. All these factors can be crucial in order to simulate the activity of catalysts grafted onto silica with different silanol densities, especially the one containing ca. 1 OH per nm², which can serve as a model for the SiO₂ surface pretreated under high vacuum and at 700 °C.

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Introduction

Amorphous silica is a widely used support in heterogeneous catalysis.^{1–3} It can be prepared by flame pyrolysis resulting in nanoparticles with a surface area of 50–380 m² g^{−1}^{4–6} or by sol-gel processes, which can provide mesostructured silica with controlled porosity and very high surface areas up to 1000 m² g^{−1}.⁷ In all cases, the bulk of silica is composed of SiO₄ tetrahedral units that form siloxane rings of different Si–O sizes. The size of the silica rings present on the silica surface generally ranges from flexible 12-membered rings to strained 4-member Si–O rings and the distribution of such siloxane rings generally depends on the calcination/activation temperature of silica. At the surface, different kinds of silanols can be found; they can be classified as isolated, geminal, vicinal and interacting silanols, which show distinct IR signatures (see Fig. 1).^{8–10} Isolated and geminal silanols appear as sharp OH bands at 3747 cm^{−1}, and vicinal silanols within the range 3750–3740 cm^{−1}. Interacting silanols, which correspond to two silanols in close proximity interacting via hydrogen bonding, appear in the IR spectrum being significantly red-shifted in the

case of hydrogen-donor silanols.¹¹ Using fumed Aerosil-200 (200 m² g^{−1}) as an example, the decrease of the surface silanol concentration is nearly exponential with increasing temperature (Fig. 1b). The ≡Si–OH decreases from ca. 3 ≡Si–OH nm^{−2} to ca. 0.8 ≡Si–OH nm^{−2} for Aerosil-200 partially dehydroxylated at 200 and 700 °C. For SiO₂ calcined at 200 °C all the three types of silanols can be found whereas for silica dehydroxylated at

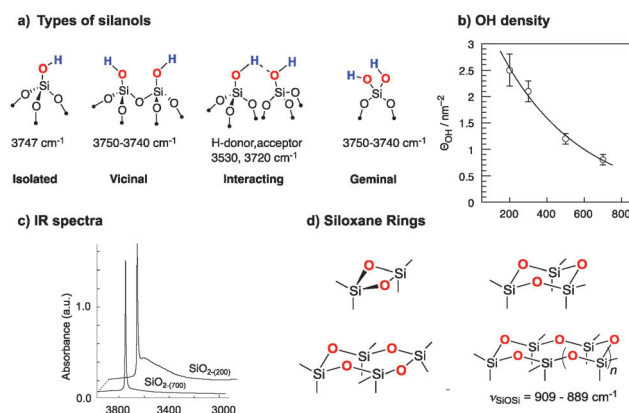


Fig. 1 (a) Types of silanols. (b) Effect of temperature on surface-OH. (c) Infrared spectra of silica partially dehydroxylated at 200 °C and 700 °C (SiO₂-(200) and SiO₂-(700)). (d) Siloxane Rings. Adapted from Wischert, *et al.*³

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700 °C (SiO_{2-700}) there are 90% of isolated silanols and 10% of geminal silanols.^{12,13} The surface can also reorganize as recently found during the grafting of organotin species.¹⁴ Another effect of the dehydroxylation temperature is that more strained siloxane bridges are formed such as 6- and 4-membered Si–O rings.

Highly dehydroxylated silica is commonly used as a support for the grafting of catalysts onto the silica surface,^{15,16} for the generation of single-site catalysts^{17–20} or the deposition of metallic nanoparticles.^{21,22}

The main difficulty in the modelling of SiO_2 surfaces arises from the amorphous character of this support. Usually, either cluster or periodic models are used when modelling silica surfaces and catalytic reactions occurring on them.^{23–25} Cluster models composed of siloxane cages with different Si–O ring sizes are commonly used in the modelling of silica surfaces when studying the reactivity of single site catalysts grafted onto isolated, geminal or vicinal silanols.^{26–30} Nevertheless, these models do not usually account for the heterogeneity of sites found on the silica and the strain energy present in highly dehydroxylated silica surfaces. Obvious alternatives to cluster models are periodic models.^{23,31–35} Parrinello and co-workers constructed an amorphous fully dehydroxylated SiO_2 surface based on molecular dynamics simulations³³ containing highly strained two-membered (2MR) Si–O rings. Tielens and co-workers generated an amorphous model containing 26 SiO_2 units,³⁶ which was terminated with OH groups,³² having an OH density equal to 5.8 OH per nm^2 and a thickness of 5–8 Å. Ugliengo *et al.* created models of amorphous SiO_2 surfaces with a broad range of silanol densities from 7.2 to 1.5 OH per nm^2 .³⁵ Removal of water molecules by condensing adjacent silanol pairs gave surfaces containing silanol densities as low as 1.5 $\equiv\text{Si}-\text{OH nm}^{-2}$. More recently, Ewing *et al.* constructed amorphous models of silica with a silanol coverage as low as 2.0 OH per nm^2 .³⁴ Also, “crystalline” periodic models based on different terminations of β -cristobalite³⁷ have been used in order to create dehydroxylated SiO_2 surfaces,³⁸ which included the direct condensation of silanols and reconstructions of the silica surface.³⁹

Therefore, although several *ab initio* models have been proposed in the literature, they still present some limitations. From one side, cluster models are generally too ideal in the sense that since they have no boundary constraints and they are more relaxed, *i.e.* they contain less strain than partially and highly dehydroxylated amorphous silica. In addition, the *ab initio* based periodic models either crystalline or amorphous are not heterogeneous enough mostly due to their limitation in size. All these aspects might be crucial when modelling single-site catalysts grafted onto highly dehydroxylated silica. In this contribution, amorphous SiO_2 models of $2.1 \times 2.1 \text{ nm}$ and silanol densities between 7.2 and 1.1 OH per nm^2 are obtained *via* direct desorption of water molecules and silica reconstruction steps. Such reconstructions are necessary in order to reach silanol coverages as low as *ca.* 1.0 OH per nm^2 and the resulting model can be used to simulate the SiO_{2-700} °C surface, a widely used system in catalysis, for which a realistic model has not been proposed yet. In addition, the dehydroxylation process described in this contribution is significantly more favoured energetically than for previously proposed amorphous models.³⁵

By using *ab initio* atomistic thermodynamics the most stable surface as a function of temperature and $P_{\text{H}_2\text{O}}$ is obtained. Both the predicted coverage under the given conditions and the IR signatures of the related surfaces are in good agreement with the experiment. Hence, the proposed models can serve as a basis to evaluate catalytic processes occurring on silica surfaces with different degrees of hydroxylation.^{3,6,12}

Construction of the model and methodology

An amorphous SiO_2 bulk cubic unit with a lattice parameter equal to 21.39 Å and containing 216 SiO_2 units (648 atoms) was taken as the starting point.⁴⁰ This structure corresponds to a $3 \times 3 \times 3$ unit cell of β -cristobalite annealed at high temperatures using classical molecular dynamics and subsequently cooled down to room temperature, as similarly described by Huff *et al.*⁴¹ This structure was optimized at the DFT level (*vide infra*) and the final configuration is shown in the ESI† (Fig. S1). Subsequently, the 001 plane of SiO_2 was taken by cleaving 10 Å of the silica bulk material in this direction. This surface termination was taken because it is the simplest choice among the low index terminations of silica. It allows taking a thicker layer of SiO_2 at lower computational cost than for the 101 and 111 terminations. The selection of only one of the three surface terminations is justified in the case of an amorphous model of silica, whereas it is more critical for the ideal terminations of β -cristobalite.³⁹ For the latter, the fully hydroxylated 001, 101 and 111 terminations give rise to very clear and distinct silanol patterns which are different from each other, containing only geminal, vicinal and isolated silanols, respectively, (see Fig. S7–S9 from the ESI†). Analogous to the 001 termination, the 101 and 111 terminations of the amorphous SiO_2 bulk structure were also constructed and optimized (see Fig. S5 and S6 of the ESI†). In contrast to the ideal terminations of β -cristobalite, the three terminations of the amorphous SiO_2 bulk model contain all kinds of silanols, and thus the differences among them are less significant (see Fig. S4–S6, ESI†).

For all the terminations, the dangling Si and O bonds on both sides of the surface were saturated with OH and H atoms, and in the case of 001 a surface as thick as 13.7 Å was obtained. Then, a vacuum slab of *ca.* 20 Å was added with a final unit cell in the *z* direction equal to 34.2 Å, large enough to avoid interactions between images in the *z* direction. The resulting surface had a unit cell equal to $21.39 \times 21.39 \times 34.2 \text{ Å}$. This fully hydroxylated SiO_2 surface (SiO_{2-1} model) contained 402 atoms with a silanol density equal to 7.2 OH per nm^2 on both the top and the bottom layers. The top and side views of this silica surface are given in Fig. 2. Also a smaller silica surface was constructed with dimensions $14.32 \times 14.32 \times 34.78 \text{ Å}$ in order to evaluate the effect of decreasing the cell size (see Fig. S2 for the corresponding bulk structure, ESI†).

All the calculations reported in this study were performed by means of the CP2K code with the revised version of the PBE functional,^{42,43} using GTH pseudopotentials,^{44,45} plane waves and Gaussian basis sets. The DZVP basis set was used for Si, O and H atoms,⁴⁶ and an energy cutoff equal to 300 Ry was selected.

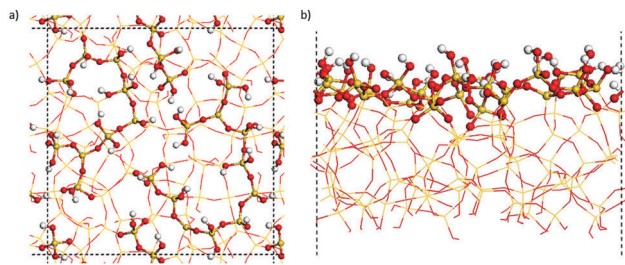


Fig. 2 Top and side views of the amorphous silica model with an initial OH density equal to 7.2 OH per nm².

Dispersion effects were considered for some selected structures by means of the DFT-D3 correction by Grimme and co-workers.⁴⁷ Since the dispersion effects were not found to be significant, these results are reported in the ESI.[†] The total energy was converged to 5×10^{-7} Hartree. The convergence criteria equal to 2×10^{-4} Hartree/bohr and 3×10^{-3} bohr were used for the gradient and the displacement, respectively. The surface phase diagram was constructed using the *ab initio* atomistic thermodynamic approach described by Reuter and Scheffler.⁴⁸

The Gibbs free energy (ΔG) was estimated using the following expression:

$$\Delta G = E_{\text{SiO}_2\text{-fully}} + mE_{\text{SiO}_2\text{-bulk}} - E_{\text{SiO}_2\text{-surf}} - n_{\text{H}_2\text{O}}\mu_{\text{H}_2\text{O}}$$

where $E_{\text{SiO}_2\text{-fully}}$ is the energy of the fully hydroxylated SiO₂ surface ($\rho = 7.2$ OH per nm²), $E_{\text{SiO}_2\text{-surf}}$ is the energy of the corresponding silica surface, $E_{\text{SiO}_2\text{-bulk}}$ is the energy of the SiO₂ unit from the bulk, m is the number of SiO₂ migration units, and $n_{\text{H}_2\text{O}}$ and $\mu_{\text{H}_2\text{O}}$ are the number of water molecules released and the chemical potential of water, respectively.

The chemical potential of water can be described by the ideal gas law, which then enables to relate a given chemical potential ($\mu_{\text{H}_2\text{O}}$) to specific temperatures (T) and pressures of H₂O ($P_{\text{H}_2\text{O}}$):

$$\mu_{\text{H}_2\text{O}}(T, P_{\text{H}_2\text{O}}) = E_{\text{H}_2\text{O}} + \mu_{\text{H}_2\text{O}}(T, P^0) + k_{\text{B}}T \ln(P_{\text{H}_2\text{O}}/P^0)$$

where $E_{\text{H}_2\text{O}}$ is the electronic energy of the water molecule, $\mu_{\text{H}_2\text{O}}(T, P^0)$ is the standard chemical potential at temperature T , which includes all the contributions from the vibrations and rotations of the molecule, and the ideal gas entropy at 1 atm. Although the standard chemical potential can be calculated from first principles calculations, the corresponding values of $\mu_{\text{H}_2\text{O}}(T, P^0)$ from JANAF thermodynamic tables were used instead.⁴⁹

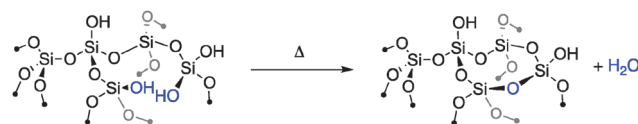
By using this approach, the phonon modes are assumed not to change upon the dehydroxylation process. While this is not inherently true, it has been previously shown for the dehydroxylation process of β -cristobalite surfaces that the resulting error is expected to be less than the errors inherent to DFT calculations.³⁹ The IR spectrum of all surfaces was obtained by using the vibrational analysis module of CP2K. Frequency calculations were performed at the harmonic level with the increment used to construct the Hessian with a finite difference method set to 0.001 bohr. Gaussian broadening was applied in order to obtain the spectra. In order to distinguish the O–H stretching modes of the top of the SiO₂ surface from the ones of the bottom layer, deuterium atoms were used for the latter.

Results and discussion

The top layer of the SiO₂ surface was dehydroxylated in successive steps. This process forms new siloxane bridges (Si–O–Si bonds) and every step is accompanied by the formation of a water molecule (Scheme 1). The energetics of the condensing nine adjacent pairs of silanol groups is reported in Table 1. Silica surfaces with silanol densities ranging between 7.2 and 3.3 OH per nm² were obtained. In all the cases, the systems were fully optimized and during these dehydration steps no defects (e.g., under- or over-coordinated Si or O atoms) were formed on the silica surface. In Table 1 the structures with their silanol coverage and relative dehydration energies are shown.

The results exemplify the heterogeneity of the new siloxane bonds being formed, with dehydration energies ranging 1–322 kJ mol^{−1}. The heterogeneity of the new siloxane bridges formed is somewhat lower in the case of the smaller unit cell, with dehydroxylation energies ranging from −2 to 282 kJ mol^{−1}.

We can compare these results with the ones previously reported for the 001 ideal termination of β -cristobalite, where dehydration energies had an energetic cost equal to 43–60 kJ mol^{−1} per water molecule depending on the dehydroxylation pattern for a decrease in the silanol coverage from 7.2 to 3.7 OH per nm². Another difference between the 001 termination of the current amorphous model and the crystalline one is that the former has all kinds of silanol groups whereas the latter contains only geminal silanols (see Fig. S7, ESI[†]). The energetics reported for the current amorphous model are also quite different when compared to the ones reported by Van Santen and co-workers



Scheme 1 Direct dehydration of two silanol groups. A new siloxane bridge is formed and one water molecule is released. The atoms involved in the dehydration reaction are depicted in blue.

Table 1 Density of OH per nm², total number of water molecules released every step, average distance of the two new Si–O bonds created upon formation of the siloxane bridges (in Å), the relative reaction energy (ΔE_r) per water molecule (in kJ mol^{−1}) and the average ΔE over all the water molecules removed with respect to the initial fully hydroxylated silica surface

Model	OH per nm ²	Total $n_{\text{H}_2\text{O}}$ released	Average distance new O–Si bonds (Å)	ΔE_r (kJ mol ^{−1})	ΔE per $n_{\text{H}_2\text{O}}$ (kJ mol ^{−1})
SiO ₂ -1	7.2	0	—	0	0
SiO ₂ -1-II	6.8	1	1.689	138	138
SiO ₂ -1-III	6.3	2	1.654	38	88
SiO ₂ -2	5.9	3	1.712	128	101
SiO ₂ -2-II	5.5	4	1.658	1	76
SiO ₂ -2-III	5.0	5	1.654	35	68
SiO ₂ -3	4.6	6	1.749	317	110
SiO ₂ -3-II	4.2	7	1.740	322	139
SiO ₂ -3-III	3.7	8	1.699	149	141
SiO ₂ -4	3.3	9	1.691	138	141

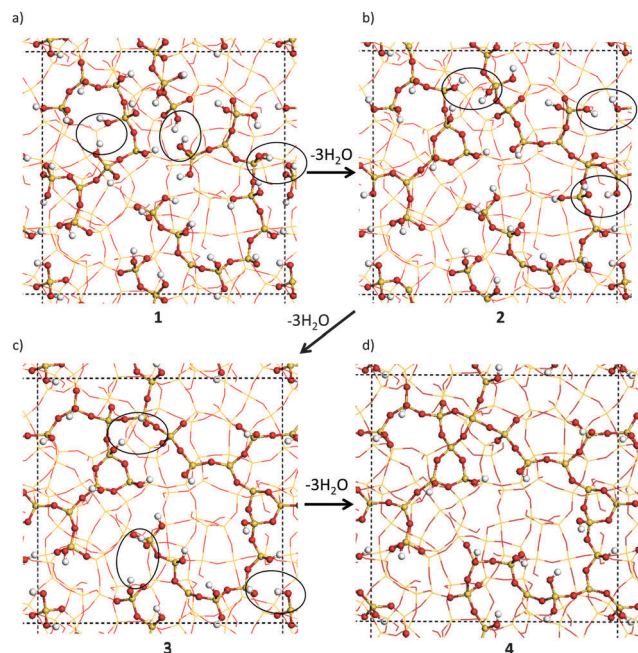


Fig. 3 Silica surfaces obtained from successive dehydration steps starting from the fully hydroxylated SiO_2 surface (a) obtaining surfaces with silanol densities equal to 5.9 OH per nm^2 (b), 4.6 OH per nm^2 (c) and 3.3 OH per nm^2 (d). The silanol groups removed after every step are depicted in circles.

for the condensation of small neutral silica units; for example, the condensation of two $\text{Si}(\text{OH})_4$ units was endothermic by only 9 kJ mol^{-1} .⁵⁰ Another difference between the currently used amorphous model and the crystalline 101 and 111 ideal surfaces of β -cristobalite is that direct dehydration is not possible on the β -cristobalite terminations, whereas direct dehydration routes can be envisaged for the 101 and 111 terminations of the amorphous model (see the ESI† for the optimized structures of these surface terminations).

The first set of three water molecules removed were selected on a geometrical basis; the silanol pairs removed were involved in short hydrogen bonds, with the $\text{SiOH} \cdots \text{OHSi}$ distances equal to 1.864, 1.976 and 2.103 Å. The removal of these three water molecules has an average energetic cost of 101 kJ mol^{-1} per water molecule, and a silica surface with OH coverage equal to 5.9 OH per nm^2 is obtained (model $\text{SiO}_2\text{-2}$ from Table 1 and Fig. 3b). The subsequent removal of three additional water molecules also involves silanol groups interacting via hydrogen bonds characterized by $\text{SiOH} \cdots \text{OHSi}$ distances equal to 1.747, 2.600 and 3.161 Å.

This step presents similar energetics since the formation of the new three siloxane bridges is in this case endothermic by 110 kJ mol^{-1} per water molecule. The resulting surface has a silanol coverage equal to 4.6 OH per nm^2 , which is typical for a silica surface treated under vacuum conditions at 180–200 °C.^{12,51} The structure of the resulting surface is shown in Fig. 3c. The last three direct dehydration reactions also involve silanol groups forming hydrogen-bonds with $\text{Si-OH} \cdots \text{OHSi}$ distances equal to 2.049, 2.721 and 3.897 Å. As expected, when the silanol density is further decreased the dehydration process

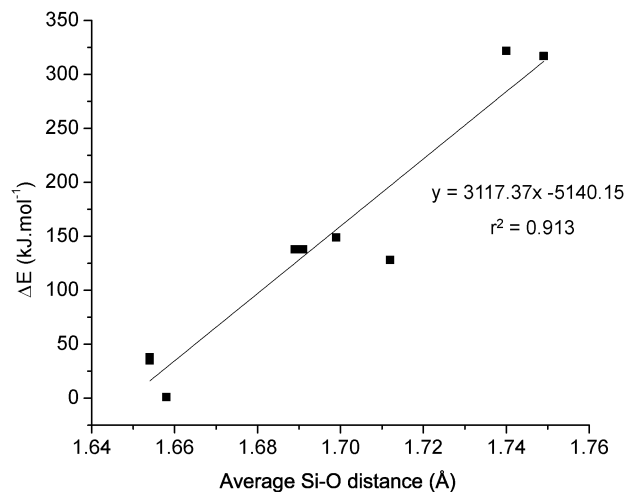
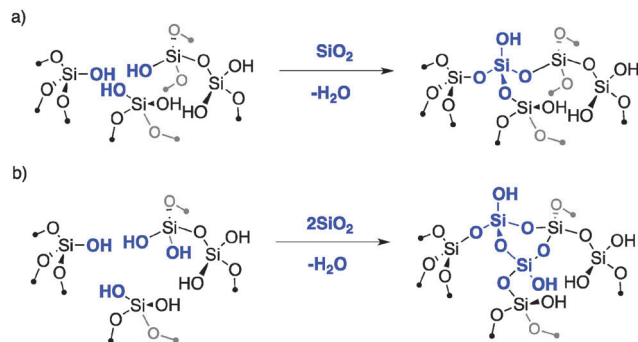


Fig. 4 Relationship between the energetic cost (ΔE , in kJ mol^{-1}) of dehydration and the average distance of the two new Si-O bonds, which are created upon formation of a new siloxane bridge in each dehydration step.

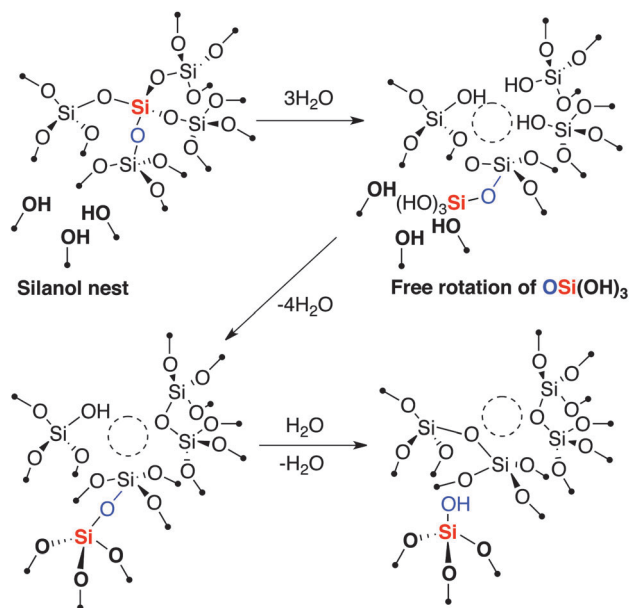
becomes more energy demanding; overall, the removal of nine water molecules to reach a silanol coverage equal to 3.3 OH per nm^2 is endothermic by 141 kJ mol^{-1} per water molecule (see Table 1), and the resulting silica surface presents the silanol coverage of $\text{SiO}_{2-(200)}$, i.e. the same compared to fumed silica under thermal treatment at 200 °C. The resulting geometries of the new siloxane bridges formed show that there is a relationship between the average Si-O distance of the siloxane bridges created upon condensation and their corresponding energetic cost (as seen in Fig. 4). Hence, it seems that the average Si-O distance can serve as a descriptor of the resulting strain introduced on the surface upon the dehydroxylation process. In the case of the smaller unit cell, the dehydration energies are higher than for the same value of averaged Si-O distances. This suggests that for the big unit cell the strain is better distributed along the cell (see Fig. S10 of the ESI†).

The removal of two additional water molecules was found to be very energy demanding in this case: 426 and 441 kJ mol^{-1} , respectively. If other Si-O-Si bonds were formed, surface sites with under- or over-coordinated Si and O atoms were formed. This is probably an indication that the surface with a silanol density equal to 3.3 OH per nm^2 already contained a high degree of strain. Thus, in order to further dehydroxylate the surface an alternative route is needed. For this purpose, the migration of SiO_2 groups was considered. This approach was recently used by Rozanska *et al.* in order to dehydroxylate ideal terminations of β -cristobalite.³⁹ Moreover, such SiO_2 reconstructions have indeed been reported experimentally for silica surfaces, which can occur via successive water-assisted Si-O bond ruptures and formations.^{12,52} Two kinds of SiO_2 reconstructions have been studied in this contribution (Scheme 2). In the first type of reconstruction, formally one SiO_2 unit (from the silica bulk unit cell) is added to three Si-OH groups in a process that releases one water molecule (Scheme 2a).

An example of how this reconstruction can take place on the silica surface is depicted in Scheme 3. When a $-\text{Si}(\text{OH})_3$ unit is



Scheme 2 Dehydroxylation of the SiO_2 surface via the migration of one (a) or two (b) SiO_2 groups. The atoms involved in the surface migration reaction are depicted in blue.



Scheme 3 example of how SiO_2 reconstruction can take place via successive water-assisted rupture and formation of siloxane bridges.

formed upon addition of three water molecules, this unit can freely rotate and upon further condensation reactions and one final water-assisted siloxane rupture, the silanol nest with three silanols can be converted to a $\text{Si}(\text{OH})(\text{OSi})_3$ unit. Finally, the migration of one SiO_2 group and the net release of one water molecule occur. This approach allows closing silanol nests without having to consider multiple water-assisted processes of rupture and formation of siloxane bonds on the silica surface.

In the second type of SiO_2 reconstruction evaluated, two SiO_2 groups react with silanol groups and one water molecule is generated (Scheme 2b). The energy per SiO_2 unit was calculated from the bulk of amorphous silica, which contains 216 SiO_2 groups in a $21.39 \times 21.39 \times 21.39 \text{ \AA}$ unit cell. Overall, the subsequent migration steps were considered (eqn (1)–(3)):

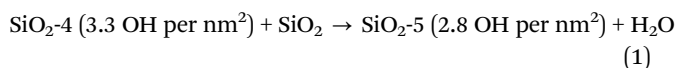
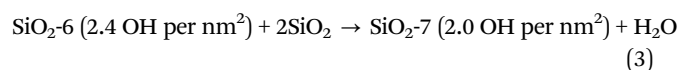
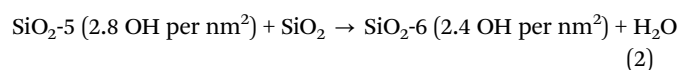


Table 2 Density of OH per nm^2 , total number of water molecules released every step, the relative reaction energy (ΔE_r) per water molecule (in kJ mol^{-1}) and the average over all the water molecules removed with respect to the initial fully hydroxylated silica surface

Model	OH per nm^2	Total $n\text{H}_2\text{O}$ released	ΔE_r (kJ mol^{-1})	ΔE per $n\text{H}_2\text{O}$ (kJ mol^{-1})
$\text{SiO}_2\text{-5}$	2.8	10	43	131
$\text{SiO}_2\text{-6}$	2.4	11	157	133
$\text{SiO}_2\text{-7}$	2.0	12	27	125
$\text{SiO}_2\text{-8}$	1.6	13	252	134
$\text{SiO}_2\text{-9}$	1.1	14	259	143



The energetic cost of every step leading to the $\text{SiO}_2\text{-7}$ model from the $\text{SiO}_2\text{-4}$ surface is reported in Table 2, and the corresponding structures ($\text{SiO}_2\text{-4-7}$) are depicted in Fig. 5. The migration of one silica unit to close a silanol nest of three silanol groups is expected to be more feasible when the three silanol groups are disposed in a symmetrical way, *i.e.* with the oxygen atoms forming the silanol nest equidistant to each other, since the resulting silanol would be less strained. This condition is better fulfilled for the silanol nest selected for the formation of the $\text{SiO}_2\text{-5}$ surface from the $\text{SiO}_2\text{-4}$ surface with respect to another possibility (see Fig. S11 from the ESI†). This reconstruction step (eqn (1)) has an energetic cost equal to 43 kJ mol^{-1} and releases one water molecule. The same type of reconstruction (eqn (2)) is also possible on the $\text{SiO}_2\text{-5}$ surface since the group of three silanols is close to each other to form the $\text{SiO}_2\text{-6}$ surface. However, in this case the reconstruction is more energy demanding: 157 kJ mol^{-1} . In the case of the $\text{SiO}_2\text{-6}$ surface, there are no three silanol groups that are close to each other in a triangular shape. Conversely, in one area of the silica surface there are four silanol units in pairs, forming a rectangular pattern (see Fig. S12, ESI†). This pattern is the one where the migration of two SiO_2 units is possible and hence it was considered to further dehydroxylate the surface. This last step releases one water molecule (eqn (3)) and it is endothermic by only 27 kJ mol^{-1} .

The obtained energetics of the dehydroxylation process shows that at this low silanol coverage (2.8 OH per nm^2), the migration of SiO_2 groups is significantly less demanding ($27\text{--}157 \text{ kJ mol}^{-1}$) than the direct condensation of adjacent silanol groups ($> 400 \text{ kJ mol}^{-1}$) since the latter process probably forms too strained Si–O–Si bonds.

Hence, by considering the silica surface reconstruction *via* the migration of SiO_2 groups, the silica surface can be further dehydroxylated with favourable energetics and a surface with a silanol density equal to 2.0 OH per nm^2 ($\text{SiO}_2\text{-7}$ model) is obtained. Finally, from the latter silica surface two successive direct dehydration steps allowed obtaining silica surfaces with silanol densities equal to 1.5 and 1.1 OH per nm^2 ($\text{SiO}_2\text{-8}$ and $\text{SiO}_2\text{-9}$ models from Fig. 5, respectively). These steps have an

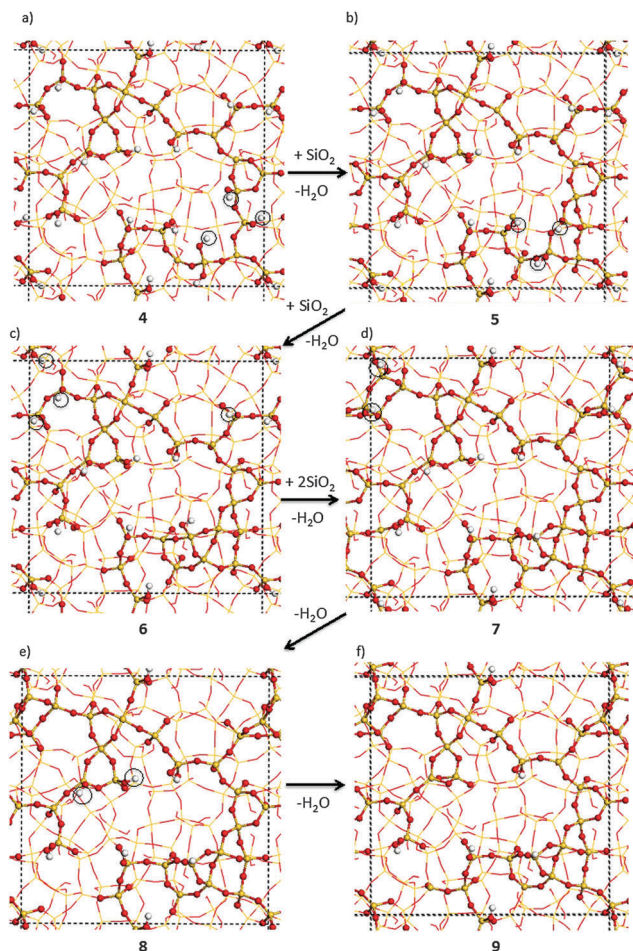


Fig. 5 Silica surfaces obtained from SiO_2 migration and direct dehydration steps starting from the SiO_2 -4 surface (3.3 OH per nm^2 , (a)) in order to obtain SiO_2 surfaces 5–9, with silanol densities equal to 2.8, (b), 2.4, (c), 2.0 (d), 1.6, (e) and 1.1 OH per nm^2 (f). The silanol groups removed after every step are depicted in circles.

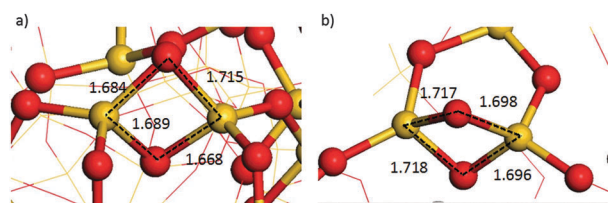


Fig. 6 2MR rings formed upon dehydration of the SiO_2 -7 surface. (a) 2MR formed upon formation of the SiO_2 -8 surface and (b) upon formation of the SiO_2 -9 surface. Distances are given in Å.

energetic cost equal to 252 and 259 kJ mol^{-1} (see Table 2), respectively, and lead to the formation of the SiO_2 -9 surface. This is probably due to the fact that during these steps two 2MR are formed (see Fig. 6 for a closer look of these strained 2MR siloxane rings).

In this specific case, the two pairs of Si–OH groups are too far from each other to consider the migration with two SiO_2 units and hence direct condensation was considered. Parrinello and co-workers also found the formation of two-membered siloxane rings on their fully dehydroxylated SiO_2 model.³³

Also, Chizallet and Raybaud reported these kinds of sites for their fully dehydroxylated model of amorphous silica–alumina.⁵³

The global energetics reported in this study to reach a silanol coverage of 1.1 OH per nm^2 presents an average energetic cost of 143 kJ mol^{-1} per water molecule. In the case of the smaller unit cell, the average energetic cost to reach the same silanol coverage is similar (145 kJ mol^{-1} per water molecule, see Table S2, ESI†). This energetics is significantly more favored than the one previously reported by Ugliengo *et al.* in their periodic amorphous model. In that case, in order to reach a silanol coverage equal to 1.5 OH per nm^2 , 262 kJ mol^{-1} per water molecule were required.

Surface phase diagram and computed IR spectra

With the energetics of the dehydroxylation steps previously described, a stability diagram including the nine silica surfaces was constructed by using the *ab initio* atomistic thermodynamic approach. The preferred hydroxyl coverage as a function of temperature (T , in K) and $\log(P_{\text{H}_2\text{O}}/P^0)$ is depicted in Fig. 7. We can observe how for this termination of our amorphous silica model (001), the starting OH coverage (7.2 OH per nm^2) is the most stable phase under standard conditions. This termination is similar to the one previously reported by Rozanska *et al.* for the fully hydroxylated 001 surface of β -cristobalite.³⁹ When increasing the temperature up to 400 K and decreasing the pressure below 10^{-5} bar, the surface with a silanol density equal to 4.6 OH per nm^2 (SiO_2 -3) becomes the most stable one. This is in agreement with the results of Zhuralev, who reported this as the preferred coverage for various kinds of silica following activation under vacuum at 180–200 °C.¹² The next most stable surface is the one containing a silanol density equal to 2.0 OH per nm^2 , which is stable over a broad range of temperatures and pressures. Indeed, this has also been observed experimentally for silica samples treated between 400 and 500 °C; the silanol density barely changed within these temperatures (*ca.* 2.0 and 1.75 OH per nm^2 , respectively).¹² Finally, the surface phase diagram predicts that at high temperatures (1000–1100 K) and low $P_{\text{H}_2\text{O}}$ pressures (*ca.* 10^{-5} – 10^{-3} bar), the most stable surface is the one containing the lowest silanol density (1.1 OH per nm^2). Hence, the most highly dehydroxylated surface, *e.g.* with *ca.* 1.0 OH per nm^2 , is the most stable phase for silica dehydroxylated under vacuum (10^{-8} bar) and at high temperatures (700 °C), commonly referred to as SiO_2 -700 °C. In order to check the validity of the silica models found along this study, the IR spectrum corresponding to the top layer of the silica surface was calculated. The results are shown in Fig. 7b–e for the silica surfaces with silanol densities equal to 7.2, 4.6, 2.0 and 1.1 OH per nm^2 , respectively. For the most hydroxylated surface the highest band of the IR spectrum is centered at 3735 cm^{-1} , corresponding to isolated silanol groups. In this spectrum there are also additional bands in the 3700 – 3400 cm^{-1} region (Fig. 7b) corresponding to interacting silanols. The analogous IR spectrum of the fully hydroxylated SiO_2 surface corresponding to a smaller unit cell is quite comparable to this one (see Fig. S13, ESI†). This IR spectrum can also be compared with the ones computed for the fully hydroxylated 001, 101 and 111 terminations of β -cristobalite proposed by Rozanska *et al.*³⁹ Indeed, the current model reproduces the

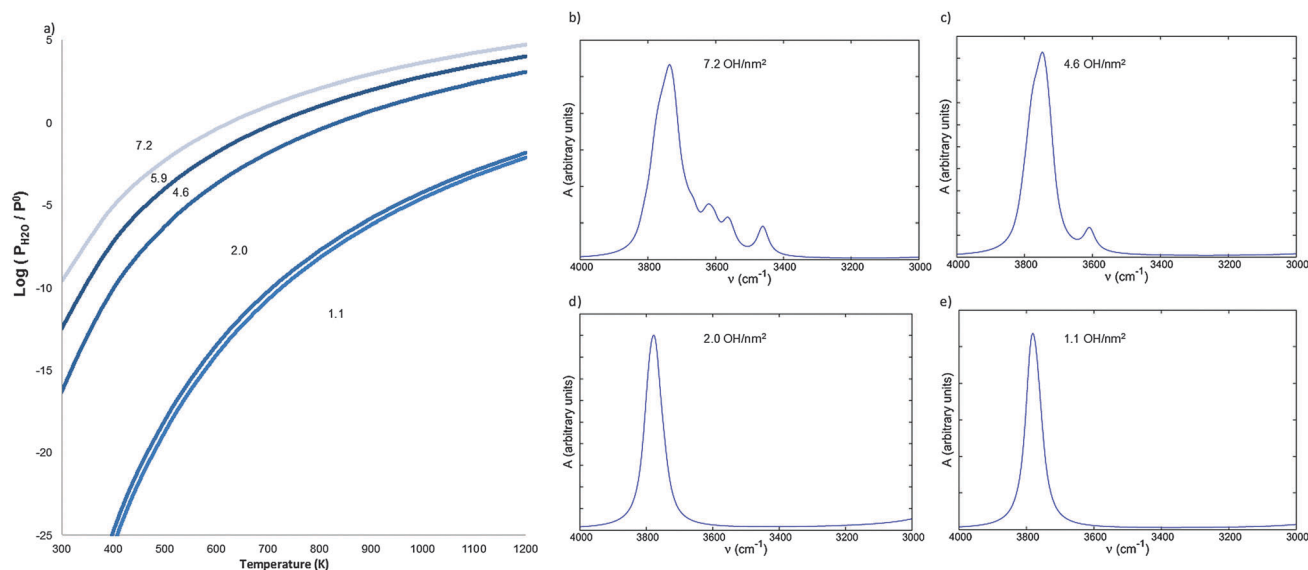


Fig. 7 (a) Surface phase diagram as a function of $\log(P_{\text{H}_2\text{O}}/P^0)$ and temperature (in K) including all the silica surfaces. For every area the most stable silanol density (in OH per nm^2) is depicted. The resulting IR spectra for selected silanol coverages. (b) 7.2, (c) 4.6, (d) 2.0 and (e) 1.1 OH per nm^2 .

experimental IR signal better due to the broader variability of the silanol groups with respect to the fully hydroxylated ideal terminations of β -cristobalite. (Fig. S14–S16, ESI†). At 4.6 OH per nm^2 , the highest band is centred at *ca.* 3750 cm^{-1} and there is only one additional single peak at *ca.* 3610 cm^{-1} , corresponding to interacting silanols (Fig. 7c). Finally, the IR spectra from the surfaces with silanol densities equal to 2.0 and 1.1 OH per nm^2 show only one single sharp band centred at *ca.* 3780 cm^{-1} (Fig. 7d–e). In this case, this signal corresponds only to isolated silanol groups. All these spectroscopic results are in good agreement with the experimentally reported IR spectra of silica at different OH coverages.

Conclusions

Amorphous silica surfaces of *ca.* $2 \times 2\text{ nm}$ with silanol densities ranging 1.1–7.2 OH per nm^2 were constructed. Starting from a fully hydroxylated SiO_2 surface (7.2 OH per nm^2), the condensation of adjacent silanol groups has an average energetic cost equal to 141 kJ mol^{-1} (in electronic energy) per water molecule until a silanol density equal to 3.3 OH per nm^2 is reached. Upon increasing the temperature and decreasing the pressure some of the silica surfaces with intermediate coverages (*e.g.* the ones equal to 5.9 and 4.6 OH per nm^2) are the most stable surfaces when taking the chemical potential of water into consideration, meaning that under such conditions their formation energies are negative in Gibbs free energy. In these dehydration steps a clear trend was observed between the resulting energetics and the average distance of the newly formed Si–O bonds. Thus, it was found that the Si–O distances serve as a descriptor of the strain introduced on the surface upon dehydroxylation. The silica models have a broad heterogeneity of siloxane bridges, as evidenced by the wide range of dehydration energies obtained: $1\text{--}322\text{ kJ mol}^{-1}$. Subsequently, surface reconstruction steps involving the migration of SiO_2 groups were considered in order to further dehydroxylate the silica

surface. Using this approach, silica surfaces with silanol densities within 3.3 and 2.0 OH per nm^2 and without under- or over-coordinated Si and O atoms were created. Finally, by condensing two additional adjacent silanol groups a highly dehydroxylated silica surface was obtained. This final model contains a hydroxyl coverage of *ca.* 1 OH per nm^2 and the average energetic cost of dehydroxylation is equal to 143 kJ mol^{-1} per water molecule. This highly dehydroxylated silica surface is the most stable one at 700°C and under high vacuum conditions, as predicted by the surface phase diagram. This is in agreement with the experimental silanol density (0.8 OH per nm^2) obtained for a silica surface treated under high vacuum at 700°C . The corresponding vibrational spectra of the Si–OH region obtained for some selected surfaces are also in qualitative good agreement with the experimental IR spectra reported for silica. All the obtained surfaces can be used in order to model a wide variety of heterogeneous systems such as silica-supported metallic nanoparticles, grafted organometallic catalysts or single-site catalysts present on the surface of silica pre-treated under different conditions. The highly dehydroxylated silica surfaces present a high degree of strain that could have strong influence on the catalytic processes occurring on them.

Notes

The coordinates for the amorphous silica models (SiO_2 -1–9) can be downloaded as ESI.†

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